Chapter 13B
Appendix: Diagnostic, Procedural, Nutrition, and Antibiotic Skills

Medvale Internal Medicine - Final Capstone Skills Segment

Purpose and Learning Frame

This appendix chapter turns scattered practical skills into a structured bedside framework. The organ-system chapters tell the learner what diseases exist; this chapter focuses on how to read common diagnostic data, how to recognize dangerous support-device complications, how to choose nutrition routes, and how to begin antibiotics without losing sight of source control, cultures, toxicity, and stewardship.

The source appendix emphasized radiographic interpretation, ECG interpretation, intravenous therapy, central venous and arterial lines, nutrition support, and antibiotic therapy. This chapter preserves those domains but rewrites them into original teaching text with modern clinical framing and high-yield numbers. It is intentionally paragraph-heavy for Graph-RAG extraction while keeping the tables narrow enough for PDF readability.

Skill domain	Primary question	Do-not-miss issue
Chest radiograph	Is the film adequate, are lines/tubes safe, and is there cardiopulmonary pathology?	Missed pneumothorax, pulmonary edema, misplaced endotracheal tube or central line.
Abdominal radiograph	Is there obstruction, perforation, severe ileus, or toxic dilation?	Free intraperitoneal air, closed-loop obstruction, volvulus, toxic megacolon.
ECG	What is the rate, rhythm, axis, interval pattern, ischemia pattern, and electrolyte/drug pattern?	STEMI, unstable arrhythmia, high-grade block, hyperkalemia, prolonged QT.
Vascular access	What device is required, where should the tip be, and what complications are device-specific?	Pneumothorax, arterial puncture, CLABSI, thrombosis, air embolism, compartment syndrome.
Nutrition support	Can the gut be used, what is aspiration/refeeding risk, and when is parenteral support justified?	Refeeding syndrome, aspiration, hyperglycemia, catheter infection, inappropriate TPN.
Antibiotics	What syndrome, site, likely organisms, severity, host risk, source control need, and toxicity constraints guide therapy?	Delayed sepsis therapy, missed source control, unnecessary broad therapy, renal toxicity, C. difficile.

I. Chest Radiograph Interpretation

A chest radiograph should be read systematically before the radiologist report is used as a substitute for clinical reasoning. The first decision is whether the image is a PA/lateral study, an AP portable film, or a limited single-view study. PA and lateral films are preferred when the patient can stand; AP portable films magnify the cardiac silhouette and often under-inspire the lungs, which can falsely suggest cardiomegaly, basilar opacity, or vascular congestion. A previous film is one of the highest-yield comparison tools because chronic hyperinflation, old scarring, calcified granulomas, prior effusions, and stable cardiomegaly can be mistaken for acute pathology.

Film Quality: RIPE Before Diagnosis

The film must be checked for rotation, inspiration, penetration, and exposure before pathology is called. A technically poor film does not make disease impossible, but it lowers confidence and should change the wording of the interpretation. For example, low inspiration can crowd the pulmonary vessels and exaggerate the heart size; underpenetration can hide retrocardiac pneumonia or lower-lobe disease; severe rotation can distort mediastinal width and hilar contours.

Quality check	How to judge it	Clinical consequence if poor
Rotation	Medial clavicular heads should be roughly equidistant from the spinous processes.	Rotation can simulate mediastinal shift, hilar enlargement, or asymmetric lung lucency.
Inspiration	A good upright CXR often shows about 9-10 posterior ribs above the diaphragm.	Low volumes crowd markings and can mimic atelectasis, edema, or basilar infiltrate.
Penetration	Thoracic vertebral bodies should be faintly visible through the cardiac silhouette.	Underpenetration hides lung bases; overpenetration can make interstitial disease or edema less visible.
Exposure/field	Lung apices, costophrenic angles, and lower neck/upper abdomen should be included when relevant.	Clipped apices can miss pneumothorax; clipped bases can miss effusion or free air.

Line and Tube Position Checklist

Portable films often exist mainly to check devices. The key is to know both the desired position and the complication that should be searched for immediately after placement. The appendix source emphasized endotracheal tube, central line, and nasogastric tube positions; clinically, the same film should also be scanned for pneumothorax, new lobar collapse, malposition, and unexpected mediastinal course.

Device	Desired position	High-yield complication pattern
Endotracheal tube	Tip usually about 3-5 cm above the carina in neutral neck position.	Too deep: right mainstem intubation with left lung atelectasis. Too high: accidental extubation risk.
Right IJ/subclavian central venous catheter	Tip near lower SVC/cavoatrial junction, not deep in right atrium.	Pneumothorax, hemothorax, arterial placement, arrhythmia if intracardiac.
Left-sided central venous catheter	Should cross midline into SVC with expected venous course.	Persistent left SVC or arterial placement if abnormal course; confirm before use if uncertain.
Nasogastric/orogastric tube	Course below diaphragm with side port below GE junction; tip in stomach.	Coiling in esophagus, bronchial placement, side port above diaphragm causing aspiration risk.
PICC	Tip in lower SVC/cavoatrial region.	Tip in azygos vein, ipsilateral IJ, or too deep; thrombosis or CLABSI with prolonged use.

PA/Lateral Read: A Practical Sequence

A simple sequence prevents blind spots: compare old film, check quality, confirm devices, then read outside-in and top-down. The bones and soft tissues are not optional; rib fracture, shoulder dislocation, clavicular fracture, subcutaneous emphysema, breast shadows, and surgical clips often provide the key clinical clue. The mediastinum should be assessed for width, tracheal deviation, aortic contour, hilar enlargement, and cardiomegaly. Lung fields should be divided into upper, middle, and lower zones and compared bilaterally. The diaphragms and costophrenic angles should be checked last because small pleural effusions and free subdiaphragmatic air are commonly missed.

Finding	Pattern clue	Most useful next thought
Pulmonary edema	Bilateral perihilar/interstitial opacities, vascular congestion, Kerley B lines, effusions, enlarged cardiac silhouette.	Ask whether this is cardiogenic edema, renal failure/volume overload, ARDS, or capillary leak.
Lobar pneumonia	Focal air-space opacity, air bronchograms, silhouetting of a specific border.	Treat based on site of care and severity; consider follow-up imaging if high malignancy risk or nonresolving.
Atelectasis	Volume loss with fissure displacement, elevated hemidiaphragm, rib crowding, mediastinal pull.	Search for mucus plugging, intubation, postoperative hypoventilation, obstructing lesion.

Pneumothorax	Visible pleural line with absent peripheral vascular markings; deep sulcus sign on supine films.	If unstable or tension physiology, treat immediately rather than waiting for CT.
Pleural effusion	Blunted costophrenic angle, meniscus sign, layering fluid; posterior blunting on lateral film.	Decide if diagnostic thoracentesis is needed; unilateral, febrile, large, or atypical effusions need workup.
Hyperinflation	Flattened diaphragms, increased AP diameter, narrow vertical heart, increased retrosternal airspace.	Suggests obstructive physiology; correlate with spirometry rather than diagnosing COPD from imaging alone.

Chest Radiograph Algorithm

- Step 1: Confirm patient, date, projection, and comparison film.
- Step 2: Assess quality: rotation, inspiration, penetration, and field coverage.
- Step 3: Check devices before interpreting disease: ETT, enteric tube, central line, pacemaker/ICD, chest tube.
- Step 4: Bones and soft tissues: fractures, lytic lesions, subcutaneous emphysema, mastectomy, surgical clips.
- Step 5: Mediastinum and heart: trachea, aortic contour, hilar size, cardiac silhouette, pulmonary vasculature.
- Step 6: Lungs and pleura: compare zones, look behind heart and under diaphragms, inspect apices and costophrenic angles.
- Step 7: Decide whether the result changes immediate management or requires CT/ultrasound/repeat film.

II. Abdominal Radiographs

Plain abdominal radiographs are no longer the definitive test for many abdominal complaints, but they remain useful for bowel gas pattern, radiopaque foreign bodies, constipation burden in selected contexts, severe ileus, toxic megacolon, and suspected perforation when CT is not immediately available. A standard KUB is a supine abdominal film; an obstruction series traditionally includes upright abdominal imaging and a chest film to look for air-fluid levels and free subdiaphragmatic air.

Question	Plain film clue	Interpretive trap
Is there free air?	Subdiaphragmatic free air on upright chest/abdomen; Rigler sign on supine film.	A negative plain film does not exclude perforation; CT is more sensitive if stable.
Small bowel obstruction?	Central dilated small bowel loops, valvulae conniventes crossing entire lumen, multiple air-fluid levels, paucity of colonic gas.	Ileus can mimic obstruction; transition point and closed-loop signs require CT.
Large bowel obstruction?	Peripheral colonic dilation with haustra that do not cross entire lumen; possible cecal dilation.	Cecal diameter above 10-12 cm raises perforation risk depending on duration and clinical context.
Ileus?	Diffuse bowel gas throughout small and large bowel, scattered loops, no clear transition.	Localized sentinel loop may indicate adjacent inflammation such as pancreatitis or cholecystitis.
Toxic megacolon?	Dilated colon, loss of haustra, systemic toxicity in IBD or infectious colitis.	Do not treat as simple constipation; avoid colonoscopy during unstable acute dilation.

III. ECG Interpretation

The ECG should be read with the same discipline every time: calibration, rate, rhythm, axis, intervals, atrial and ventricular enlargement, QRS morphology, ischemia, repolarization, and comparison with old tracings. The most common error is jumping to ST segments before checking rate, QRS width, baseline, lead placement, and prior ECG. A second common error is describing a rhythm without asking whether the patient is stable; unstable tachyarrhythmia or bradyarrhythmia is a management problem first and a diagnostic puzzle second.

Measurement	Normal / key threshold	High-yield interpretation
Paper calibration	25 mm/s, 10 mm/mV standard.	At standard speed, 1 small box = 0.04 s, 1 large box = 0.20 s.

Rate	300 / large boxes between R waves if regular; 10-second strip x 6 if irregular.	Tachycardia >100/min; bradycardia <60/min.
PR interval	120-200 ms.	Long PR = first-degree AV block; progressively lengthening then dropped QRS = Mobitz I; fixed PR with dropped QRS = Mobitz II.
QRS duration	<120 ms.	Wide QRS suggests bundle branch block, ventricular rhythm, paced rhythm, hyperkalemia, or sodium channel blockade.
QTc	Often concerning if >450 ms in men, >470 ms in women; markedly high risk often >500 ms.	Torsades risk rises with QT-prolonging drugs, hypokalemia, hypomagnesemia, bradycardia, congenital syndromes.
ST elevation	Contiguous leads and clinical setting determine meaning.	STEMI, pericarditis, early repolarization, LVH, LBBB/paced rhythm, aneurysm can all elevate ST segments.

Rate, Rhythm, Axis

Rate is mechanical, rhythm is relational, and axis is directional. First ask whether the QRS complexes are narrow or wide, regular or irregular, and whether P waves are present before each QRS. Sinus rhythm requires upright P waves in lead II with a consistent PR interval and one P wave for each QRS. Axis is estimated from leads I and aVF: positive in both is normal; positive I/negative aVF is left axis deviation; negative I/positive aVF is right axis deviation; negative in both suggests extreme axis deviation or lead reversal.

ECG pattern	Likely diagnosis	Next clinical question
Irregularly irregular narrow-complex rhythm without consistent P waves.	Atrial fibrillation.	Is the patient unstable, is rate control needed, and is anticoagulation indicated?
Regular narrow-complex tachycardia around 150/min with flutter waves.	Atrial flutter with 2:1 conduction.	Look in inferior leads and V1; consider AV nodal blockade or cardioversion depending on stability.
Regular narrow-complex tachycardia 150-250/min with hidden/retrograde P waves.	SVT/AVNRT/AVRT.	Try vagal maneuvers/adenosine if stable and no contraindication.
Wide-complex regular tachycardia.	Ventricular tachycardia until proven otherwise.	Treat as VT when uncertain, especially with structural heart disease or instability.
Progressive PR lengthening then dropped QRS.	Mobitz I.	Often nodal; assess symptoms, medications, inferior MI, vagal tone.
Dropped QRS without progressive PR prolongation.	Mobitz II/high-grade block.	Higher risk infranodal disease; consider pacing evaluation.
No relationship between P waves and QRS complexes.	Complete heart block.	Assess escape rhythm, hemodynamics, reversible causes, pacing need.

Ischemia, Infarction, and ST-T Patterns

ECG ischemia interpretation is anatomical. Contiguous lead changes are more persuasive than isolated abnormalities. Inferior leads are II, III, and aVF; lateral leads include I, aVL, V5, and V6; anterior/septal leads include V1-V4 depending on distribution. Reciprocal ST depression increases concern for acute coronary occlusion. Posterior MI may present as horizontal ST depression and tall R waves in V1-V3; right ventricular MI should be suspected with inferior MI, hypotension, clear lungs, and elevated JVP.

Lead territory	Typical leads	Associated artery pattern
Inferior	II, III, aVF	RCA more common; LCx possible. Check right-sided leads if hypotension/RV infarct suspected.
Septal/anterior	V1-V4	LAD territory; extensive anterior MI has

		higher risk of cardiogenic shock and LV dysfunction.
Lateral	I, aVL, V5-V6	LCx or diagonal branch; may accompany anterior or inferior MI.
Posterior	ST depression V1-V3, tall R waves; posterior leads V7-V9 confirm.	Often LCx or RCA; do not dismiss as nonspecific ST depression if clinical picture fits.
Diffuse concave ST elevation with PR depression	Widespread leads, often except aVR/V1.	Pericarditis pattern; distinguish from STEMI by distribution, reciprocal changes, symptoms, troponin, echo.

Electrolyte and Drug Patterns

Abnormality	ECG clues	Why it matters
Hyperkalemia	Peaked T waves, PR prolongation, QRS widening, sine-wave pattern in severe cases.	Can rapidly deteriorate into ventricular arrhythmia; calcium stabilizes myocardium when ECG changes are present.
Hypokalemia	ST depression, T-wave flattening, prominent U waves, apparent QT/QU prolongation.	Predisposes to ventricular arrhythmias and digoxin toxicity.
Hypercalcemia	Short QT interval.	Often malignancy or hyperparathyroidism; severe cases can cause arrhythmia.
Hypocalcemia	Prolonged QT interval.	Torsades risk, especially with other QT-prolonging factors.
Digoxin effect/toxicity	Scooped ST depression; toxicity may cause atrial tachycardia with block, PVCs, AV block, bidirectional VT.	Check renal function, potassium, interacting drugs, level timing, and symptoms.
Sodium channel blockade	Wide QRS, terminal R in aVR, hypotension/seizures in TCA overdose.	Sodium bicarbonate is treatment for severe TCA/sodium-channel blocker toxicity.

IV. Intravenous Therapy and Vascular Access

Vascular access is a risk-benefit decision. The question is not simply whether a line can be placed; the question is what access meets the clinical need with the lowest complication burden. Peripheral IVs are preferred for routine short-term fluids and medications. Midlines and PICCs serve intermediate or prolonged access needs but carry thrombosis and infection risks. Non-tunneled central lines are appropriate for unstable patients, vasopressors, high-osmolality infusions, emergent dialysis access, or rapid central access, but they should be removed as soon as they are no longer needed.

Access type	Typical use	Major limitations/complications
Peripheral IV	Short-term fluids, common IV medications, contrast, blood products in many cases.	Infiltration, phlebitis, extravasation injury; not ideal for vesicants or prolonged high-osmolality infusions.
Midline	Intermediate-duration peripheral-compatible therapy.	Tip is not central; do not use for true central-only infusions.
PICC	Weeks to months of IV antibiotics, chemotherapy in selected cases, difficult access.	Upper-extremity DVT, CLABSI, malposition; not ideal in advanced CKD if future dialysis access needed.
Non-tunneled CVC	ICU access, vasopressors, emergent access, TPN, central irritants, hemodynamic monitoring.	Pneumothorax, arterial puncture, arrhythmia, CLABSI, thrombosis, air embolism.
Tunneled catheter/port	Long-term chemotherapy, chronic infusions, long-term parenteral nutrition.	Procedure-related risk, infection, thrombosis; port requires needle access.

Arterial line	Continuous BP monitoring, frequent ABGs, unstable vasopressor patients.	Bleeding, thrombosis, limb ischemia, infection, inaccurate waveform if improperly leveled.

Peripheral IV Complications

Peripheral IV complications should be treated early because minor findings can become limb-threatening when irritant or vesicant medications extravasate. Phlebitis causes redness, tenderness, warmth, and a palpable cord along the vein. Infiltration produces swelling and coolness from nonvesicant fluid leakage. Extravasation is medication leakage that can injure tissue; severe pain, blistering, skin necrosis, neurovascular compromise, or tight compartments require urgent escalation.

Problem	Clue	Immediate response
Phlebitis	Red, tender vein; pain along catheter path.	Remove IV, warm compress, assess need for new site and infection evaluation if purulent/systemic signs.
Infiltration	Swelling, cool skin, slowed infusion; nonvesicant fluid.	Stop infusion, remove/replace IV, elevate limb, monitor compartment symptoms.
Extravasation	Pain, swelling, blistering, tissue injury from vesicant/irritant.	Stop infusion, leave catheter initially if antidote/aspiration needed, follow drug-specific protocol, urgent consult if severe.
Air embolism	Dyspnea, chest pain, hypoxia, hypotension after line manipulation.	Clamp line, left lateral/Trendelenburg positioning may be used, high-flow O2, urgent critical care.

Central Line Safety

After central venous catheter placement, the clinician should confirm tip position and search for pneumothorax or hemothorax when the insertion site carries that risk. The catheter should be necessary every day it remains in place. Infection prevention depends on hand hygiene, maximal sterile barriers during insertion, chlorhexidine skin antisepsis, avoidance of femoral site when feasible in adults, meticulous hub care, dressing care, and prompt removal when no longer needed. A central line placed only for routine blood draws is generally not justified.

Complication	Mechanism / clue	Prevention or response
Pneumothorax	Pleural injury during subclavian/IJ placement; dyspnea or apical pleural line.	Use ultrasound when appropriate, confirm after placement; chest tube if large/symptomatic/tension.
Arterial puncture	Bright pulsatile blood, pressure waveform, abnormal course on imaging.	Do not dilate uncertain vessel; confirm with ultrasound/pressure/transduction.
Arrhythmia	Guidewire/catheter irritates myocardium.	Withdraw wire/catheter if ectopy occurs; monitor during placement.
CLABSI	Fever/bacteremia without another source; line tenderness or purulence may be absent.	Culture appropriately, remove if indicated, empiric therapy based on severity and organism risk.
Thrombosis	Swelling, pain, catheter dysfunction, venous obstruction.	Use smallest necessary device/lumens, remove unnecessary line, image when suspected.
Air embolism during removal	Air entrainment through open venous tract.	Supine/Trendelenburg if tolerated, occlusive dressing, patient exhales/holds breath as appropriate.

V. Nutrition Support

Nutrition support begins with the question: can the gut be used safely? Enteral nutrition is preferred over parenteral nutrition when the gastrointestinal tract is functional because it supports gut integrity, avoids central-line dependence, and has fewer infectious and metabolic complications. Parenteral nutrition is not a more complete or safer form of nutrition; it is a rescue strategy when enteral intake is impossible, contraindicated, or inadequate for a clinically meaningful period.

Route	Best use	Important complications
Oral diet + supplements	Functional swallowing and gut, mild to moderate nutritional risk.	Aspiration if dysphagia is missed; inadequate intake if monitoring is poor.
Nasogastric/orogastric tube	Short-term gastric feeding or decompression.	Sinusitis, aspiration, malposition, nasal trauma; confirm placement before use.
Nasojejunal/post-pyloric tube	High aspiration risk or gastric intolerance in selected patients.	Placement difficulty, clogging, still not zero aspiration risk.
PEG/G-tube	Longer-term enteral access when stomach route is appropriate.	Peristomal infection, leakage, aspiration, buried bumper, inappropriate use in advanced terminal illness.
Jejunostomy tube	Long-term post-pyloric feeding when gastric feeding unsuitable.	Clogging, diarrhea, surgical/procedural risk.
Parenteral nutrition	Nonfunctional GI tract, prolonged inability to meet needs enterally, severe malabsorption, short bowel, bowel obstruction/ileus in selected cases.	Hyperglycemia, electrolyte shifts, refeeding, hepatic dysfunction, catheter infection, thrombosis.

Refeeding Syndrome and Monitoring

Refeeding syndrome is a predictable danger when calories are restarted in a severely malnourished patient. Insulin release drives phosphate, potassium, and magnesium into cells. The classic laboratory pattern is hypophosphatemia, often accompanied by hypokalemia, hypomagnesemia, fluid retention, arrhythmias, weakness, respiratory failure, and neurologic changes. High-risk patients should receive cautious calories, thiamine, electrolyte repletion, and frequent monitoring during the first days of refeeding.

High-risk clue	Why it matters	Practical response
Very low intake for days to weeks, alcoholism, eating disorder, cancer cachexia, severe weight loss.	Total body phosphate, magnesium, potassium, and thiamine stores may be depleted.	Check baseline electrolytes, give thiamine, start calories cautiously, monitor daily or more often if unstable.
Low phosphate after feeds start.	Hallmark biochemical clue; can cause respiratory muscle weakness and arrhythmia.	Replete phosphate and slow advancement of calories.
Fluid overload during nutrition initiation.	Sodium/water retention can worsen heart failure or pulmonary edema.	Use careful fluid strategy and daily weights; adjust PN/EN volume.

VI. Antibacterial Antibiotic Therapy

Antibiotic therapy should be syndrome-based, severity-based, and source-control-based. The most important early step in sepsis is timely active therapy, but the most important stewardship step after stabilization is reassessment. Cultures, imaging, drainage, debridement, device removal, susceptibilities, renal function, allergy history, and clinical trajectory should narrow therapy. Broad-spectrum antibiotics are not a diagnosis; they are a temporary bridge while the diagnosis is clarified.

Decision point	Clinical question	RAG-friendly principle
Syndrome	Pneumonia, UTI, cellulitis, meningitis, intra-abdominal infection, endocarditis, neutropenic fever?	The same drug can be right or wrong depending on site penetration and organisms.

Host risk	Immunosuppression, recent hospitalization, prior resistant organism, dialysis, structural lung/urinary disease?	Risk factors change empiric coverage more than fever alone.
Severity	Sepsis, shock, meningitis, necrotizing infection, endocarditis, bacteremia?	Severe disease lowers tolerance for delayed active therapy.
Source control	Abscess, infected catheter, obstruction, empyema, necrotic tissue, infected stone?	Antibiotics often fail without drainage/removal/debridement.
Toxicity constraints	Kidney disease, QT prolongation, pregnancy, myasthenia, C. difficile history, drug interactions?	The safest effective narrow regimen is preferred once data arrive.
Reassessment	Cultures, susceptibilities, clinical response, stop date.	De-escalation and defined duration are active management, not undertreatment.

Antibiotic Class Quick Reference

Class	Common coverage/use pattern	Signature cautions
Penicillins	Streptococci, enterococci for ampicillin/amoxicillin, anaerobes with beta-lactamase inhibitor combinations; antipseudomonal options include piperacillin-tazobactam.	Allergy history matters; rash is not always IgE allergy. Adjust for renal function when needed.
Cephalosporins	Generation-dependent: cefazolin for MSSA/surgical prophylaxis; ceftriaxone for many CAP/GN infections; cefepime for Pseudomonas risk.	Limited enterococcus coverage; cefepime neurotoxicity risk in renal failure.
Carbapenems	Broad gram-negative, anaerobic, ESBL coverage; severe polymicrobial infections.	Overuse selects resistance; ertapenem lacks Pseudomonas coverage.
Vancomycin	MRSA, serious gram-positive infections, severe C. difficile orally.	Nephrotoxicity, infusion reaction; monitor dosing/exposure for serious infections.
Macrolides	Atypical pneumonia coverage, pertussis, some respiratory syndromes.	QT prolongation, drug interactions; resistance limits monotherapy in many settings.
Doxycycline/tetracyclines	Atypicals, tick-borne disease, acne, CA-MRSA in some skin infections.	Photosensitivity, esophagitis; avoid in pregnancy and young children in classic teaching though doxy exceptions exist in rickettsial disease.
Fluoroquinolones	Respiratory or gram-negative coverage depending on agent; urinary and intra-abdominal uses in selected patients.	Tendinopathy, neuropathy, QT prolongation, dysglycemia, aortic risk warnings, C. difficile; reserve when alternatives poor.
Aminoglycosides	Severe gram-negative synergy or selected resistant infections.	Nephrotoxicity and ototoxicity; monitor levels and renal function.
TMP-SMX	UTI, Pneumocystis, some CA-MRSA, Nocardia.	Hyperkalemia, renal effects, rash/SJS, cytopenias, interactions with warfarin.
Metronidazole	Anaerobes below diaphragm, bacterial vaginosis, trichomoniasis, C. difficile alternatives in older regimens.	Neuropathy with prolonged use, metallic taste, disulfiram-like warning historically emphasized.
Clindamycin	Anaerobes above diaphragm, toxin suppression adjunct in necrotizing strep/clostridial disease, some skin infections.	High C. difficile risk; resistance varies.
Linezolid/daptomycin	Resistant gram-positive infections; linezolid for MRSA pneumonia, daptomycin for bacteremia/right-sided endocarditis.	Linezolid: cytopenias/serotonergic interactions. Daptomycin: myopathy, not for pneumonia.

Empiric Therapy Logic by Syndrome

Syndrome	Initial empiric logic	Immediate reassessment trigger
Community pneumonia	Cover typical and atypical respiratory pathogens based on outpatient/inpatient/severity and comorbidity	No improvement, hypoxia, effusion, resistant organism risk, alternative diagnosis.

	risk.	
Cellulitis	Nonpurulent: streptococci; purulent/abscess: drainage plus MRSA consideration.	Necrotizing pain out of proportion, bullae, crepitus, systemic toxicity.
UTI/pyelonephritis	Base on cystitis vs pyelo, pregnancy, resistance, obstruction, prior cultures.	Sepsis, flank pain with obstruction, persistent fever, male/complicated anatomy.
Intra-abdominal infection	Enteric gram-negatives and anaerobes; source control is central.	Peritonitis, abscess, perforation, biliary obstruction, ischemic bowel.
Meningitis	Rapid bactericidal CNS-penetrating therapy plus adjuncts when indicated.	Do not delay therapy for LP if CT/instability delays procedure.
Catheter bloodstream infection	Assess severity, line necessity, organism risk; cover MRSA and gram-negative risk if unstable.	S. aureus, Candida, persistent bacteremia, tunnel infection usually push toward removal.

VII. Integrated Skill Algorithms

Algorithm: Unstable Patient With a New Portable CXR

- First: confirm airway device position if intubated. A right mainstem ETT is a fixable cause of sudden hypoxemia.
- Second: look for pneumothorax, especially after line placement or with sudden ventilator pressure change.
- Third: check pulmonary edema, aspiration, large effusion, lobar collapse, and new infiltrate.
- Fourth: verify central line/enteric tube/chest tube position before using or troubleshooting the device.
- Fifth: compare with prior imaging and decide whether ultrasound or CT is needed after stabilization.

Algorithm: ECG With Chest Pain

- Confirm ECG quality and compare with old ECG if immediately available.
- Identify STEMI-equivalent patterns, contiguous ST elevation, reciprocal changes, posterior MI pattern, or new concerning conduction abnormality.
- If ECG is nondiagnostic but suspicion remains high, repeat ECGs and obtain troponins; early occlusion can evolve.
- Search for mimics: pericarditis, LVH strain, LBBB/paced rhythm, early repolarization, hyperkalemia, pulmonary embolism.
- Treat the patient, not just the tracing: instability, persistent pain, shock, or malignant arrhythmia changes urgency.

Algorithm: Choosing Nutrition Route

- Can the patient eat safely and enough? If yes, use oral diet with supplements and monitoring.
- If oral intake is inadequate but GI tract works, use enteral feeding; choose gastric vs post-pyloric based on aspiration risk and tolerance.
- If enteral feeding is impossible, contraindicated, or persistently inadequate and the duration/risk justifies it, consider parenteral nutrition.
- Before feeding high-risk malnourished patients, address thiamine and electrolytes and monitor for refeeding syndrome.
- Reassess daily: nutrition route should change as swallowing, gut function, procedures, and goals of care change.

VIII. Study Tables: Numbers Worth Remembering

Number / threshold	Context	Meaning
3-5 cm above carina	Endotracheal tube tip	Common practical acceptable target on CXR in neutral position.
9-10 posterior ribs	CXR inspiration	Approximate sign of adequate inspiration on upright PA film.
Cardiothoracic ratio <50%	PA chest film	Heart width less than half thoracic width; AP films exaggerate size.
PR 120-200 ms	ECG interval	Long PR suggests first-degree AV block; pattern of dropped beats differentiates

		Mobitz types.
QRS <120 ms	ECG interval	Wide QRS changes tachycardia differential and raises conduction/toxin/electrolyte concerns.
QTc >500 ms	ECG risk threshold	Marked torsades risk, especially with drugs/electrolyte abnormalities.
Cecum >10-12 cm	Large bowel dilation	Higher perforation risk depending on duration and cause.
Daily line necessity	Central line stewardship	Remove unnecessary central lines to reduce infection and thrombosis.
First 3-5 days of refeeding	Nutrition monitoring	Highest-risk window for electrolyte shifts, especially phosphate.

References and Source Basis

- Uploaded source: appendix.pdf, pages covering chest radiograph interpretation, abdominal radiographs, ECG interpretation, IV therapy, central venous/arterial lines, nutritional support, and antibacterial antibiotic therapy.
- American College of Radiology Appropriateness Criteria: imaging selection principles for acute chest pain and acute respiratory illness.
- CDC Core Elements of Hospital Antibiotic Stewardship Programs: stewardship framework for appropriate antibiotic use.
- ASPEN clinical guidance and nutrition support literature: enteral/parenteral nutrition, monitoring, and prevention of complications.
- Standard ECG teaching conventions used in internal medicine and emergency medicine practice: rate, rhythm, axis, intervals, ischemia, and electrolyte/drug patterns.